

3.2 Infinite Series

Definition: A **series** (specifically infinite series) results when we want to add up all the terms of a sequence.

For example:

$\{2, 4, 6, 8, 10, \dots\}$ is a sequence.

$2 + 4 + 6 + 8 + 10 + \dots$ is a series.

Notation (Sigma Notation):

When we write $\sum_{i=1}^n a_i$, we mean $\sum_{i=1}^n a_i = a_1 + a_2 + a_3 + \dots + a_n$. (Finite series)

When we write $\sum_{i=1}^{\infty} a_i$, we mean $\sum_{i=1}^{\infty} a_i = a_1 + a_2 + a_3 + \dots + a_n + \dots$. (Infinite series)

Note: $\sum_{i=1}^n a_i$ is known as the n th partial sum of the series $\sum_{i=1}^{\infty} a_i$.

Definition: Given a series $\sum_{i=1}^{\infty} a_i$, let s_n denote its n th partial sum so that $s_n = \sum_{i=1}^n a_i$.

If the sequence $\{s_n\}$ converges to a real number S (that is $\lim_{n \rightarrow \infty} s_n = S$), then

- we say that the series $\sum_{i=1}^{\infty} a_i$ is convergent and we write $\sum_{i=1}^{\infty} a_i = s$.
- The number S is called the sum of the series.

If the sequence $\{s_n\}$ is divergent, then the series is called divergent.

Example 1: Find the first, second, third, and fourth partial sum of the following series.

$$1-1+1-1+1-1+\dots$$

Does the series converge or diverge?

$$s_1 = 1$$

$$s_2 = 1-1=0$$

$$s_3 = 1-1+1=1$$

$$s_4 = 1-1+1-1=0$$

We see that the sequence of partial sum $\{s_n\} = \{1, 0, 1, 0, \dots\}$ which clearly does not converge.

Therefore the given series diverges.

Example 2: Find the first, second, third, and fourth partial sums of the series: $\sum_{i=1}^{\infty} \left(\frac{1}{2}\right)^i$

We see that:

$$s_1 = \sum_{i=1}^1 \left(\frac{1}{2}\right)^i = \frac{1}{2}$$

$$s_2 = \sum_{i=1}^2 \left(\frac{1}{2}\right)^i = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$s_3 = \sum_{i=1}^3 \left(\frac{1}{2}\right)^i = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{7}{8}$$

$$s_4 = \sum_{i=1}^4 \left(\frac{1}{2}\right)^i = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} = \frac{15}{16}$$

Note: It appears that the series in the above example converges to 1.

Example 3: Show that the series $\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$ is convergent, and find its sum.

We will use the partial fraction decomposition that we learned in chapter 3 on $\frac{1}{k(k+1)}$ (why we are doing this will become clear in a moment).

In particular we seek real numbers A and B so that $\frac{1}{k(k+1)} = \frac{A}{k} + \frac{B}{k+1}$. If we multiply both sides of this equation by $k(k+1)$, we obtain $1 = A(k+1) + Bk \Leftrightarrow 1 = (A+B)k + A$. From this equation we see that $A=1$ and $B=-1$.

So, we can rewrite the series as $\sum_{k=1}^{\infty} \frac{1}{k(k+1)} = \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+1} \right)$. Moreover the n^{th} partial sum of the series is: $s_n = \sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} \right) = \left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1} \right) = 1 - \frac{1}{n+1}$.

Note: Quite awesomely, many of the terms of this partial sum cancel out, and we are left with a quite simple formula for the partial sum of the series. Mathematicians call this a “telescoping” series.

Now, using the ideas of section 5.1 we have that $\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1} \right) = 1 - \left(\lim_{n \rightarrow \infty} \frac{1}{n+1} \right) = 1$.

Thus, this series is convergent by definition and $\sum_{k=1}^{\infty} \frac{1}{k(k+1)} = 1$.

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Definition: A series is called a geometric series if it is of the form $\sum_{i=1}^{\infty} ar^{i-1} = a + ar + ar^2 + ar^3 + \cdots$ where $a \neq 0$. Each term in such a series is obtained from the previous one by multiplying it by a common ratio r .

For example: The series $\sum_{i=1}^{\infty} \left(\frac{1}{2} \right)^i$ is a geometric series with $a = \frac{1}{2}$ and $r = \frac{1}{2}$.

Theorem: The geometric series $\sum_{i=1}^{\infty} ar^{i-1} = a + ar + ar^2 + ar^3 + \dots$ is

- convergent if $|r| < 1$ and its sum is $\sum_{i=1}^{\infty} ar^{i-1} = \frac{a}{1-r}$.
- divergent if $|r| \geq 1$

Proof:

We first note that if $r = 1$, then $s_n = \sum_{i=1}^n a(1^{i-1}) = na$, and since $a \neq 0$ $\lim_{n \rightarrow \infty} s_n$ is positive or negative infinity. This implies $\sum_{i=1}^{\infty} ar^{i-1}$ is divergent when $r = 1$.

Now, suppose $r \neq 1$. We note that in this case $s_n = \sum_{i=1}^n ar^{i-1}$ and $rs_n = \sum_{i=1}^{\infty} ar^i$. These two equations imply

$$rs_n - s_n = \sum_{i=1}^n ar^i - \sum_{i=1}^n ar^{i-1} = (ar + ar^2 + \dots + ar^{n-1} + ar^n - (a + ar + ar^2 + \dots + ar^{n-1})) = ar^n - a.$$

From this we obtain: $s_n(r-1) = a(r^n - 1)$. Now, since $r \neq 1$, we have $s_n = \frac{a(r^n - 1)}{r - 1}$. Now, we will examine $\lim_{n \rightarrow \infty} s_n$ in two cases.

- If $|r| > 1$ or $r = -1$, we have $\lim_{n \rightarrow \infty} s_n = \frac{a(\lim_{n \rightarrow \infty} (r^n) - 1)}{r - 1}$. We know from section 9.1 that if $|r| > 1$ or $r = -1$, then $\lim_{n \rightarrow \infty} (r^n)$ does not exist (Note: Does not exist could mean positive or negative infinity). Thus,

$\lim_{n \rightarrow \infty} s_n$ does not exist and $\sum_{i=1}^{\infty} ar^{i-1}$ is divergent.

- If $|r| < 1$, we know from section 9.1 that $\lim_{n \rightarrow \infty} (r^n) = 0$, so $\lim_{n \rightarrow \infty} s_n = \frac{a(\lim_{n \rightarrow \infty} (r^n) - 1)}{r - 1} = \frac{-a}{r - 1} = \frac{a}{1 - r}$. Thus,

$$\lim_{n \rightarrow \infty} s_n = \frac{a}{1 - r}, \quad \sum_{i=1}^{\infty} ar^{i-1} \text{ is convergent, and } \sum_{i=1}^{\infty} ar^{i-1} = \frac{a}{1 - r}.$$

Now, we have proven all that needed to be shown. ■

Example 4: Consider the number $0.\bar{9}$ (point 9 repeating).

a) Write this value as a geometric series in sigma notation.

We recognize that $0.\bar{9} = 0.9 + 0.09 + 0.009 + \dots = \frac{9}{10} + \frac{9}{100} + \frac{9}{1000} + \dots$

Thus, we can represent point 9 repeating as a geometric series with $a = \frac{9}{10}$ and $r = \frac{1}{10}$. This means

$$0.\bar{9} = \sum_{i=1}^{\infty} \left[\frac{9}{10} \left(\frac{1}{10} \right)^{i-1} \right].$$

b) Find the exact value of the series from part (a).

This is a geometric series with $a = \frac{9}{10}$ and $r = \frac{1}{10}$. Since $|r| < 1$, we know by the above theorem that

the series is convergent and: $\sum_{i=1}^{\infty} \left[\frac{9}{10} \left(\frac{1}{10} \right)^{i-1} \right] = \frac{9/10}{1-1/10} = \frac{9/10}{9/10} = 1$. Thus, $0.\bar{9}$ is EXACTLY the same thing as 1.

Example 5 (you try):

Determine whether the following series is convergent or divergent. If it is convergent find its exact sum:

$$\sum_{n=1}^{\infty} (8^{2n-1} \cdot 3^{3-4n})$$

With a little bit of simplification we see

$$\begin{aligned} \sum_{n=1}^{\infty} (8^{2n-1} \cdot 3^{3-4n}) &= \sum_{n=1}^{\infty} ((8^{2n})(8^{-1})(3^3)(3^{-4n})) = \sum_{n=1}^{\infty} \left(\left(\frac{64^n}{8} \right) \left(\frac{27}{81^n} \right) \right) \\ &= \sum_{n=1}^{\infty} \left(\left(\frac{27}{8} \right) \left(\frac{64}{81} \right)^n \right) = \sum_{n=1}^{\infty} \left(\left(\frac{27 \cdot 64}{8 \cdot 81} \right) \left(\frac{64}{81} \right)^{n-1} \right) = \sum_{n=1}^{\infty} \left(\left(\frac{8}{3} \right) \left(\frac{64}{81} \right)^{n-1} \right). \end{aligned}$$

So, we see that this is a geometric series with $a = \frac{8}{3}$ and $r = \frac{64}{81}$. Since $|r| < 1$, we know by the above theorem

that the series is convergent and $\sum_{n=1}^{\infty} (8^{2n-1} \cdot 3^{3-4n}) = \frac{8/3}{1-64/81} = \frac{8/3}{17/81} = \frac{81 \cdot 8}{3 \cdot 17} = \frac{216}{17}$.

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Example 6:

Consider the series $\sum_{k=1}^{\infty} \frac{1}{k}$ which is called the *harmonic series*. The n^{th} partial sum of the series is: $s_n = \sum_{k=1}^n \frac{1}{k}$.

- (a) Find s_5 (exactly).

$$s_5 = \sum_{k=1}^5 \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = \frac{137}{60} = 2.28\bar{3}.$$

This can be done on the calculator by typing in the home screen. The sum command can be found pushing the **MATH** button and then looking of the “sum” option.

- (b) Find s_{100} (rounded to four decimal places).

From the calculator, we obtain: $s_{100} \approx 5.1874$

- (c) Find s_{200} (rounded to four decimal places).

From the calculator, we obtain: $s_{200} \approx 5.8780$

- (d) Based upon your answers to parts (a) through (c), do you believe the harmonic series to be convergent or divergent? If you believe it is convergent, what do you think it converges to?

Note: The next theorem and proof answers this question rigorously.

Theorem: The harmonic series is divergent.

Proof: Let s_n be the n^{th} partial sum of the harmonic series. We begin by noting that

$$s_2 = 1 + \frac{1}{2},$$

$$s_4 = s_2 + \frac{1}{3} + \frac{1}{4} = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) > 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) = 1 + \frac{2}{2}, \text{ and}$$

$$s_8 = s_4 + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) > 1 + \frac{2}{2} + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) = 1 + \frac{3}{2}.$$

If we continue this pattern n times where $n > 3$ we obtain:

$$s_{2^n} = s_{2^{n-1}} + \sum_{i=2^{n-1}+1}^{2^n} \left(\frac{1}{i}\right) > 1 + \frac{n-1}{2} + \sum_{i=2^{n-1}+1}^{2^n} \left(\frac{1}{2^n}\right) = 1 + \frac{n-1}{2} + 2^{n-1} \left(\frac{1}{2^n}\right) = 1 + \frac{n-1}{2} + \frac{1}{2} = 1 + \frac{n}{2}.$$

This shows that for any positive integer N , $s_{2^N} \geq 1 + N > N$. Thus, $\lim_{n \rightarrow \infty} s_n = \infty$, and the harmonic series is divergent. ■

Note: This theorem allows one to deduce what some may consider a counterintuitive fact. Specifically, if

$\lim_{n \rightarrow \infty} a_n = 0$ *we do NOT necessarily have that* $\sum_{i=1}^{\infty} a_i$ *is convergent!!!!!!!!!!!!!!!!!!!!!!*

Note: To make the above proof more convincing one could use mathematical induction rather than just saying “if we continue this pattern...”